



Paper Id: 250839

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Subject Code: RCAIE13

Roll No:

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MCA-INT
(SEM VIII) THEORY EXAMINATION 2024-25
COMPILER DESIGN

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 07 = 14

Q no.	Question
a.	Define Finite Automata.
b.	Construct a finite automata that will accept strings of zeros and ones that contain even number of zeros and odd number of ones.
c.	What are the various phases of compilation process.
d.	What do you mean by Top Down Parsing?
e.	Define Left factoring.
f.	Write Postfix representation of the expression $(a-b)*(c+d)+(a-b)$.
g.	Define flow graph.

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

Q no.	Question
a.	Consider the following grammar, and obtain an equivalent grammar containing no useless grammar symbols. $A \rightarrow xyz \mid Xyzz$ $X \rightarrow Xz \mid xYx$ $Y \rightarrow yYy \mid XZ$ $Z \rightarrow Zy \mid z$
b.	Write short note on Lexical Analyzer Design.
c.	Discuss both types of LR Parser conflicts.
d.	What do you mean by syntax-directed translation scheme? Discuss it.
e.	What is Peephole optimization? Discuss.

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question
a.	Consider the following grammar and eliminate all the ϵ -productions without changing the language generated by the grammar: $S \rightarrow AaA$ $A \rightarrow Sb \mid bCC \mid \epsilon$ $C \rightarrow CC \mid abb$
b.	Discuss context-free grammar with example.



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4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	Discuss format of the input or source file of LEX.
b.	Write short note on Lexical Analysis phase of compilation.

5. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	Why LR Parsing is attractive? Discuss.
b.	Discuss Sentential Form Handles.

6. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	What do you mean by synthesized and inherited attributes? Discuss both of them.
b.	Write short note on representing three-address statements.

7. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	What is code optimization? Discuss.
b.	Discuss problems that hinder good code generation.

part of the following:	
Question	
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part of the following:	
Question	
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